Chapter 2. Topology of Euclidean Space

Open and closed sets, Bolzano–Weierstrass, Cauchy sequences, series, compactness, connectedness

Contents

2.1 Euclidean space **2**

2.2 Open and closed sets **2**

2.3 Bolzano–Weierstrass and Cauchy sequences **3**

2.4 Convergence tests for series **4**

2.5 Compact sets **4**

2.6 Connected sets **5**

We now work in the coordinate space $\mathbb{R}^n$. The notions of this chapter — open and closed sets, compactness, connectedness — are what make limits and continuity work in several variables.

2.1 Euclidean space

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with componentwise addition and scalar multiplication. The **inner product** and **norm** are

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i y_i, \qquad \|x\| = \sqrt{\langle x, x\rangle},$$

and the distance is $d(x, y) = \|x - y\|$.

Theorem 2.1 (Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $d(x, z) \le d(x, y) + d(y, z)$.

Proof. If $y = 0$ there is nothing to prove. Otherwise $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$ for all real t; the choice $t = \langle x, y\rangle / \|y\|^2$ gives $\langle x, y\rangle^2 \le \|x\|^2\|y\|^2$, with equality iff $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

We also use $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, which lets us pass between the Euclidean norm and coordinates. The **open ball** is $N(x, r) = \{y : \|y - x\| < r\}$.

2.2 Open and closed sets

Definition (Open and closed sets)

$U \subset \mathbb{R}^n$ is **open** if every $x \in U$ has some ball $N(x, r) \subset U$. F is **closed** if its complement is open. A point x is a **limit point** of A if every ball $N(x, r)$ contains a point of A other than x; A' denotes the set of limit points. The **closure** is $\overline{A} = A \cup A'$, the **interior** A° is the set of x with a ball inside A, and the **boundary** is $\overline{A} \setminus A^\circ$. Points of $A \setminus A'$ are **isolated**.

Open balls are open (by the triangle inequality, $N(y, r - \|y - x\|) \subset N(x, r)$).

Proposition 2.2

$\emptyset$ and $\mathbb{R}^n$ are open; arbitrary unions and finite intersections of open sets are open. Dually, arbitrary intersections and finite unions of closed sets are closed.

Proof. For a finite intersection, take the smallest of the finitely many radii. The closed statements follow by De Morgan's laws. (Infinite intersections of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.) ∎

Proposition 2.3

(a) $x \in A'$ iff there is a sequence in $A \setminus \{x\}$ converging to x; and then every ball about x contains infinitely many points of A. (b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) Pick $a_k \in N(x, 1/k) \cap A$, $a_k \neq x$. If a ball $N(x, r)$ met $A \setminus \{x\}$ in finitely many points, a smaller ball would miss them all. (b) Follows from (a), using constant sequences for points of A. ∎

Theorem 2.4

For $F \subset \mathbb{R}^n$ the following are equivalent: (a) F is closed; (b) $F' \subset F$; (c) whenever $x_k \in F$ and $x_k \to x$, then $x \in F$.

Proof. (a)⇒(b): if $x \notin F$, some ball about x misses F, so $x \notin F'$. (b)⇒(c): a limit of points of F is in F or is a limit point of F (Proposition 2.3). (c)⇒(a): if $x \notin F$ and every ball $N(x, 1/k)$ met F, we would get a sequence in F converging to $x \notin F$. ■

In particular $\overline{A}$ is the smallest closed set containing A, and A' is always closed: if x is a limit point of A', each ball about x contains a point of A', hence (by shrinking) infinitely many points of A.

A set $Y \subset X$ is **open in** X (relatively open) if $Y = X \cap U$ for some open $U \subset \mathbb{R}^n$; equivalently, every $y \in Y$ has a ball with $N(y, r) \cap X \subset Y$. Relatively closed sets are defined likewise. This is the right notion when functions are defined only on X.

2.3 Bolzano–Weierstrass and Cauchy sequences

A sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges, because of the comparison between $\|\cdot\|$ and coordinates.

Theorem 2.5 (Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point. Equivalently, every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. Enclose A in a closed box B_0 (a product of closed intervals). Bisect each side to split B_0 into 2^n boxes; one of them, B_1, contains infinitely many points of A. Repeat to get boxes $B_0 \supset B_1 \supset \cdots$, each containing infinitely many points of A, with diameters $\to 0$. Applying the nested interval theorem (Theorem 1.11) in each coordinate gives a point $x \in \bigcap B_k$, and every ball about x contains some B_k, hence infinitely many points of A. For a bounded sequence with infinitely many distinct values, choose terms closer and closer to such a limit point with increasing indices; if it has finitely many values, some value repeats infinitely often. ■

For a bounded real sequence, let A be the set of limits of its convergent subsequences (subsequential limits). Then $\limsup x_n = \max A$ and $\liminf x_n = \min A$: by Proposition 1.9, $x_n > \limsup - \epsilon$ infinitely often yields a subsequence converging to lim sup, and no subsequential limit can exceed it.

Definition (Cauchy sequence)

(x_k) in $\mathbb{R}^n$ is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_k - x_l\| < \epsilon$ for all $k, l \geq N$.

Theorem 2.6 (Cauchy criterion)

A sequence in $\mathbb{R}^n$ converges iff it is Cauchy.

Proof. Convergent sequences are Cauchy ($\|x_k - x_l\| \leq \|x_k - x\| + \|x - x_l\|$). A Cauchy sequence is bounded (take $\epsilon = 1$), so by Bolzano–Weierstrass a subsequence $x_{k_j} \to x$. Given ϵ, choose N from the Cauchy condition and j with $k_j \geq N$ and $\|x_{k_j} - x\| < \epsilon$; then $\|x_k - x\| < 2\epsilon$ for $k \geq N$. ■

Applied to partial sums: $\sum a_k$ converges iff for every $\epsilon > 0$ there is N with $\|\sum_{k=m}^{n} a_k\| < \epsilon$ for $n \geq m \geq N$. In particular absolute convergence ($\sum \|a_k\| < \infty$) implies convergence, and $a_k \to 0$ is necessary for convergence.

2.4 Convergence tests for series

Proposition 2.7 (Rearrangements)

If $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ is a bijection, then $\sum a_{\tau(n)} = \sum a_n$ (both finite or both infinite). The same holds for absolutely convergent series of reals.

Proof. Every partial sum of $\sum a_{\tau(n)}$ is at most $\sum a_n$, and conversely, so the suprema agree. For an absolutely convergent series write $a_n = a_n^+ - a_n^-$ with $a_n^\pm \geq 0$ and apply the first part to each. ■

Conditionally convergent series behave differently: by Riemann's theorem a convergent but not absolutely convergent real series can be rearranged to converge to any prescribed value (take positive terms until the target is exceeded, then negative terms, and so on; the terms tend to 0).

Theorem 2.8 (Alternating series)

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and the error after n terms is at most x_{n+1}.

Proof. The even partial sums s_{2m} increase, the odd ones s_{2m+1} decrease, $s_{2m} \leq s_{2m+1}$, and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$. Nested intervals $[s_{2m}, s_{2m+1}]$ shrink to the sum. ■

Theorem 2.9 (Comparison, root and ratio tests)

(a) If $0 \leq a_n \leq b_n$ and $\sum b_n < \infty$, then $\sum a_n < \infty$. (b) Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, it diverges. (c) If $\limsup |a_{n+1}/a_n| < 1$ the series converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large n, it diverges.

Proof. (a) Bounded partial sums. (b) If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n; compare with the geometric series. If $\alpha > 1$, $|a_n| > 1$ infinitely often, so $a_n \not\to 0$. (c) If the ratios are eventually $< \beta < 1$, then $|a_n| \leq C\beta^n$. ■

For positive c_n,

$$\liminf \frac{c_{n+1}}{c_n} \leq \liminf c_n^{1/n} \leq \limsup c_n^{1/n} \leq \limsup \frac{c_{n+1}}{c_n},$$

so the root test is stronger than the ratio test (whenever the ratio test decides, so does the root test). When $\alpha = 1$ neither test decides: $\sum 1/n$ diverges and $\sum 1/n^2$ converges.

2.5 Compact sets

Definition (Compactness)

An **open cover** of K is a family of open sets whose union contains K. K is **compact** if every open cover has a finite subcover.

Proposition 2.10

A compact set is closed and bounded. A closed subset of a compact set is compact.

Proof. The balls $N(0, k)$ cover K; a finite subcover gives boundedness. If $x \notin K$, the sets $U_k = \{y : \|y - x\| > 1/k\}$ cover K, so finitely many do, and then a ball $N(x, 1/k)$ misses K: K^c is open. For $F \subset K$ closed, add F^c to any open cover of F to get a cover of K. ■

Theorem 2.11

Every closed box $B = [a_1, b_1] \times \cdots \times [a_n, b_n]$ is compact.

Proof. Suppose an open cover has no finite subcover of B. Bisecting as in Theorem 2.5, one of the 2^n sub-boxes has no finite subcover; repeat to get nested boxes B_k with diameters $\to 0$, none finitely covered. They share a point x, which lies in some open set U of the cover; but $U \supset N(x, r) \supset B_k$ for large k, a contradiction. ■

Theorem 2.12 (Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent: (a) K is compact; (b) K is closed and bounded; (c) every sequence in K has a subsequence converging to a point of K; (d) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b) is Proposition 2.10; (b)⇒(a) since K is a closed subset of a compact box. (b)⇒(c): Bolzano–Weierstrass plus closedness (Theorem 2.4). (c)⇒(d): pick distinct points of the infinite set. (d)⇒(b): if K were unbounded, points x_k with $\|x_k\| > k$ would form an infinite set with no limit point; if some $x \in K' \setminus K$, points of K converging to x would have x as their only limit point, which is not in K. ■

Theorem 2.13 (Finite intersection property)

If $\{K_i\}$ is a family of compact sets such that every finite subfamily has nonempty intersection, then $\bigcap_i K_i \neq \emptyset$.

Proof. Fix K_1. If $\bigcap K_i = \emptyset$, the open sets K_i^c cover K_1, so finitely many do: $K_1 \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

Example 2.1

(a) A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point. Otherwise $S = \{s_1, s_2, \dots\}$ is perfect; choose closed balls $\overline{N}_1 \supset \overline{N}_2 \supset \cdots$ with centers in S, radii $\to 0$, each meeting S, and $s_k \notin \overline{N}_k$ (possible because no point is isolated). The compact sets $\overline{N}_k \cap S$ are nested and nonempty, so they share a point of S, which differs from every s_k. Hence, e.g., the Cantor set (closed, no isolated points) is uncountable.
(b) $(0, 1]$ is not compact: the cover $\{(1/k, 2) : k \in \mathbb{N}\}$ has no finite subcover.

2.6 Connected sets

Definition (Connectedness)

$S \subset \mathbb{R}^n$ is **disconnected** if there are open sets U, V with $S \subset U \cup V$, $S \cap U \neq \emptyset \neq S \cap V$ and $S \cap U \cap V = \emptyset$; otherwise S is **connected**.

Proposition 2.14

A subset of $\mathbb{R}$ is connected iff it is an interval (i.e. $x < z < y$ with $x, y \in S$ implies $z \in S$).

Proof. If S is not an interval, some $z \notin S$ lies between points of S, and $(-\infty, z)$, (z, ∞) disconnect S. Conversely let S be an interval separated by U, V, with $a \in S \cap U$, $b \in S \cap V$, say $a < b$. Let $c = \sup\{x \in [a, b] : x \in U\}$; $c \in S$. If $c \in U$, then $c < b$ and, U being open, points slightly larger

than c in $[a, b]$ lie in U, contradicting the definition of c. If $c \in V$, then $c > a$ and a neighbourhood of c lies in V, so points slightly smaller than c are not in U and some smaller number is an upper bound: contradiction. ■

Proposition 2.15

If connected sets C_i have a common point, then $\bigcup C_i$ is connected. A path-connected set (any two points joined by a continuous path in the set) is connected, and an open connected subset of $\mathbb{R}^n$ is path-connected.

Proof. If U, V separated the union, each C_i would lie entirely in U or entirely in V (by connectedness of C_i), and the common point forces all into the same one. Path-connectedness implies connectedness by Theorem 3.7 in Chapter 3 (continuous images of intervals are connected) and the first statement applied to the paths from a fixed point. For an open connected G, the set of points reachable by polygonal paths from a fixed x_0 is open, and so is its complement in G (balls are convex); connectedness forces it to be all of G. ■

Convex sets, in particular balls and boxes, are connected. The closure of a connected set is connected.